# Project Design Phase
# Proposed Solution

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| Date | 18 February 2026 |
| Team ID | LTVIP2026TMIDS81316 |
| Project Name | TransLingua:AI-Powered Multi-Language Translator |
| Maximum Marks | 2 Marks |

**Proposed Solution :**

| S.No. | Parameter | Description |
|---|---|---|
| 1. | Problem Statement (Problem to be solved) | Many individuals, students, travelers, researchers, and businesses face communication barriers due to language differences. Existing translation tools either lack contextual understanding, provide inaccurate translations, or require complex installations. There is a need for an AI-powered, easy-to-use, web-based solution that provides fast and accurate multi-language translation. |
| 2. | Idea / Solution description | TransLingua is a web-based AI-powered multi-language translator built using Streamlit and Google Gemini API. The system detects the source language automatically, translates text into the selected target language, and displays the output instantly. It provides a simple interface with language selection, swap functionality, and real-time translation. |
| 3. | Novelty / Uniqueness | • Automatic language detection feature<br>• Uses advanced Generative AI (Gemini 2.5 model) for context-aware translation<br>• Clean and lightweight web interface<br>• No installation required (browser-based)<br>• Dynamic language handling and swap option |
| 4. | Social Impact / Customer Satisfaction | The solution reduces communication gaps globally, supports students and researchers in academic work, assists travelers in foreign countries, and helps businesses expand internationally. |
| 5. | Business Model (Revenue Model) | • Freemium model (basic-free, premium-paid)<br>• API usage subscription for enterprises<br>• Advertisement-based model for free users<br>• SaaS model for organizations requiring bulk translations |
| 6. | Scalability of the Solution | The system is scalable because it uses cloud-based AI APIs. The backend can handle multiple requests simultaneously. Deployment on cloud platforms (Streamlit Cloud, AWS, or GCP) enables horizontal scaling based on user demand. |